



2026

HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

Total marks:
70

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 6 – 11)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Section I

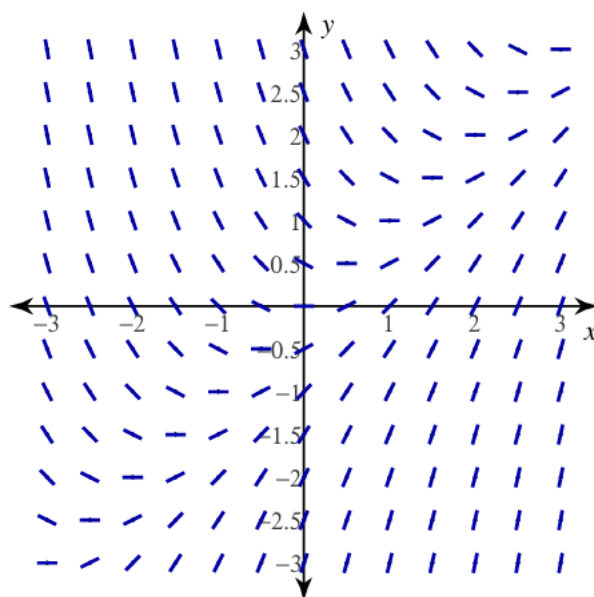
10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 to 10.

1. Consider the following slope field.



Which of the following differential equations best describes the given slope field?

- (A) $\frac{dy}{dx} = x + y$
- (B) $\frac{dy}{dx} = x - y$
- (C) $\frac{dy}{dx} = xy$
- (D) $\frac{dy}{dx} = -xy$
2. There are twelve red, nine blue and seven green socks. What is the minimum number of socks that need to be selected at random to ensure that you have a pair of socks of matching colour?
- (A) 3
- (B) 4
- (C) 8
- (D) 13

3. The vector $\underline{a} = 4\underline{i} + 3\underline{j}$ is projected onto $\underline{b} = \underline{i} - \underline{j}$.

Let $\hat{u}$ be the unit vector in the direction of this projection. Which of the following represents $\hat{u}$?

- (A) $\frac{1}{\sqrt{2}}\underline{i} - \frac{1}{\sqrt{2}}\underline{j}$
(B) $-\frac{1}{\sqrt{2}}\underline{i} + \frac{1}{\sqrt{2}}\underline{j}$
(C) $\frac{1}{\sqrt{2}}\underline{i} + \frac{1}{\sqrt{2}}\underline{j}$
(D) $-\frac{1}{\sqrt{2}}\underline{i} - \frac{1}{\sqrt{2}}\underline{j}$

4. The graph of a cubic polynomial $y = f(x)$ has the following features:

- Zeros at $x = -2, 1$ and 3
- $f(x) > 0$ for $-2 < x < 1$ and $x > 3$
- $f(x) < 0$ for $x < -2$ and $1 < x < 3$
- $f(x) \rightarrow \infty$ as $x \rightarrow \infty$

Which of the following correctly describes the graph of $y^2 = f(x)$?

- (A) Domain: $-2 \leq x \leq 1$ and $x \geq 3$; symmetric about the x -axis. Sharp cusps at $x = -2, 1$ and 3 .
(B) Domain: $-2 \leq x \leq 1$ and $x \geq 3$; symmetric about the x -axis. Vertical tangents at $x = -2, 1$ and 3 .
(C) Domain: $-2 \leq x \leq 1$ and $x \geq 3$; symmetric about the y -axis. Vertical tangents at $x = -2, 1$ and 3 .
(D) Domain: $x \leq -2$ and $1 \leq x \leq 3$; symmetric about the x -axis. Vertical tangents at $x = -2, 1$ and 3 .

5. Let $\underline{a}$ and $\underline{b}$ be two unit vectors such that $|\underline{a} - \underline{b}|^2 = 8$.

What is the value of $\underline{a} \cdot \underline{b}$?

- (A) -3
(B) 3
(C) -4
(D) 4

6. A factory produces components with a 2% defect rate. A quality inspector randomly samples components independently and tests them out one at a time.

What is the minimum number of components that must be sampled so that the probability of finding at least one defective component exceeds 0.5?

- (A) 17
(B) 34
(C) 35
(D) 50
7. If $t = \tan\left(\frac{x}{2}\right)$ and it is known that $\sin x = \frac{24}{25}$ with x in the first quadrant, what is the value of t ?
- (A) $\frac{4}{3}$
(B) $\frac{24}{25}$
(C) $\frac{12}{25}$
(D) $\frac{3}{4}$
8. Eight students are to be seated in a row. Ekun and Inupa refuse to sit next to each other. Kaedon, Ronak and Gurjivan must sit together. How many seating arrangements are possible?
- (A) 1440
(B) 2160
(C) 2880
(D) 3600

9. Which of the following represents the domain of $f(x) = \sin^{-1}\left(\frac{x^2 - 3x + 2}{x^2 + 2x + 7}\right)$?

- (A) $[1, \infty)$
- (B) $[-1, 2]$
- (C) $[-1, \infty)$
- (D) $(-\infty, 2]$

10. Which of the following represents the value of

$$\frac{1}{0!51!} + \frac{1}{1!50!} + \frac{1}{2!49!} + \dots + \frac{1}{50!1!} + \frac{1}{51!0!}?$$

- (A) $\frac{2^{51}}{50!}$
- (B) $\frac{2^{51}}{51!}$
- (C) $\frac{2^{50}}{50!}$
- (D) $\frac{2^{50}}{51!}$

End of Section I

Section II

60 marks

Attempt questions 11 -14

Allow about 1 hour 45 minutes for this section

Answer each question on the appropriate page of your answer booklet

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Answer on the appropriate page

- (a) The parametric equations of a curve is given below 2

$$x = \cos 2t$$

$$y = \cos t$$

Find the Cartesian equation of this curve.

- (b) Find the coefficient of x^4 in the expansion of $(1+2x)(1-x)^8$. 2

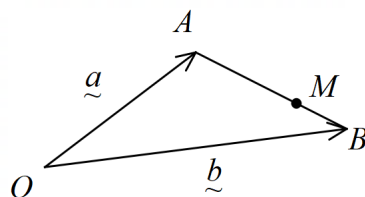
- (c) Find $\int \frac{2}{\sqrt{4-25x^2}} dx$. 2

- (d) Find the values of x for which $\frac{x+4}{x-1} \leq 6$. 3

- (e) The position vectors of three points A, B and C relative to an origin O is given by

$$\overrightarrow{OA} = a \quad \overrightarrow{OB} = b \quad \overrightarrow{OC} = 4b - 3a.$$

The point M lies on the line segment AB such that $\overrightarrow{AM} = \frac{3}{4} \overrightarrow{AB}$.



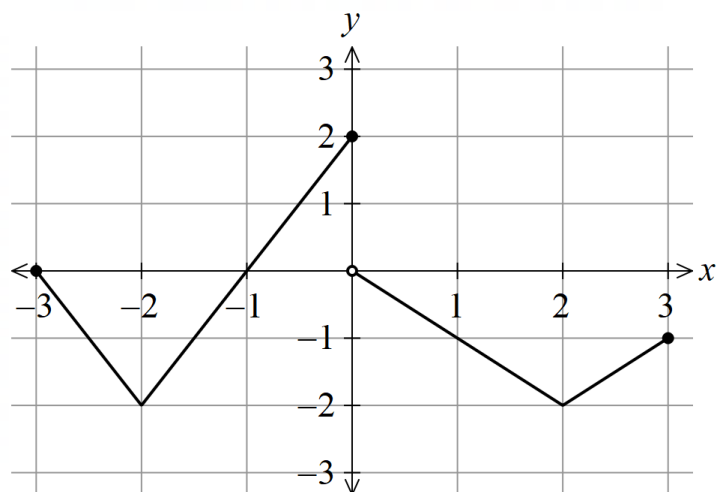
Not to scale

- (i) Prove that the points A, M and C lie on a straight line 2
- (ii) State the exact ratio of the lengths $AM : MC$. 1
- (f) Prove by mathematical induction for all positive integers that $2^n + 6^n$ is divisible by 8. 3

End of Question 11

Question 12 (15 marks) Answer on the appropriate page

(a) The graph of $y = f(x)$ is shown below.



(i) Solve the equation $|f(x)| = x$.

2

(ii) Sketch the graph of $y = \frac{1}{f(x)}$.

2

(b) By using the substitution $u = 1 + \tan^2 x$, evaluate $\int_0^{\frac{\pi}{4}} \frac{\tan x \sec^2 x}{(1 + \tan^2 x)^2} dx$.

3

(c) The polynomial $P(x) = x^3 - 5x^2 + 8x + k$ has a root of multiplicity 2 at $x = \alpha$, where k is an integer constant and α is a real constant.

(i) Find the two possible values of α .

2

(ii) Hence, determine the value of k .

2

(d) (i) Show the identity $\cos 3x \cos x \equiv \frac{1}{2}(\cos 2x + \cos 4x)$.

1

(ii) Hence, or otherwise, solve $4 \cos x \cos 2x \cos 3x = 1$ for $0 \leq x \leq \pi$.

3

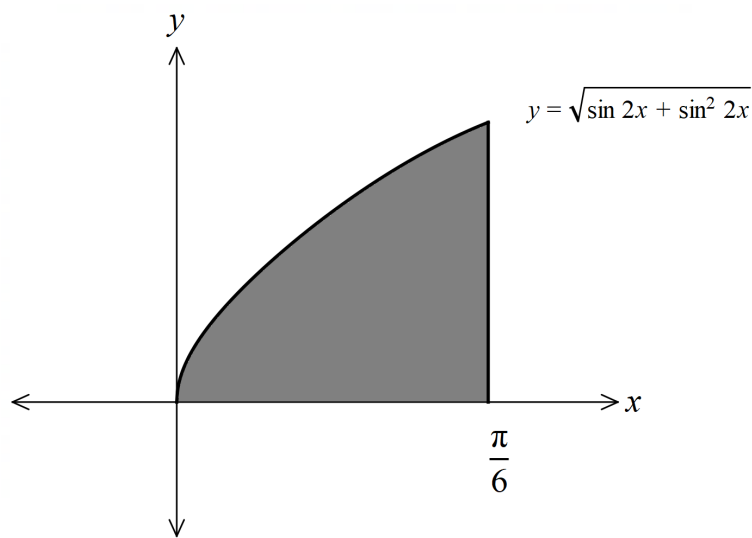
End of Question 12

Question 13 (15 marks) Answer on the appropriate page

(a) Given $\sin y = \frac{x}{1+x^2}$ for the interval $\frac{\pi}{2} < y < \pi$, show that $\frac{dy}{dx} = \frac{x^2 - 1}{(1+x^2)\sqrt{1+x^2+x^4}}$. **3**

(b) The diagram below shows the curve with equation $y = \sqrt{\sin 2x + \sin^2 2x}$ for $0 \leq x \leq \frac{\pi}{6}$. **3**

The shaded region is bounded by the curve, $x = \frac{\pi}{6}$ and the x -axis.



Find the exact volume of the solid produced when the shaded region is rotated about the x -axis.

Question 13 continues on the following page

Question 13 (continued)

- (c) Consider the vectors $\underline{u} = \underline{a} + 3\underline{b}$, $\underline{v} = 7\underline{a} - 5\underline{b}$, $\underline{w} = 7\underline{a} - 2\underline{b}$ and $\underline{z} = \underline{a} - 4\underline{b}$. 4
If $\underline{u} \perp \underline{v}$ and $\underline{z} \perp \underline{w}$, find the size of the angle between $\underline{a}$ and $\underline{b}$.

- (d) A wildlife conservation organisation releases a foundational population of 100 animals of endangered species into a protected game preserve. Ecologists estimate that the preserve has a maximum carrying capacity of 1000 animals. The herd size, H , grows over time, t years. Based on available resources, the population's rate of growth is modelled by the logistic differential equation

$$\frac{dH}{dt} = kH \left(1 - \frac{H}{1000} \right),$$

where k is a growth proportionality constant.

- (i) Use the fact $\frac{1000}{H(1000-H)} = \frac{1}{H} + \frac{1}{1000-H}$ to show that the population curve 3
takes the form $H = \frac{1000}{1 + 9e^{-kt}}$.
- (ii) Biologists track the numbers in the herd and record that the population has 2
grown to 134 animals at the end of year 2.
Determine the value of k , correct to four decimal places.

End of Question 13

Question 14 (15 marks) Answer on the appropriate page

- (a) After Mr Ribbans causes a road accident, fuel begins leaking onto a flat section of the motorway at a steady rate of $0.0008\text{m}^3/\text{s}$, forming a circular fuel spill. 2



Assuming the fuel spill forms a cylinder of constant thickness 3mm, find the exact rate at which the radius of the fuel spill is increasing, when the radius has reached 3m.

- (b) The variables x and t satisfy the differential equation 4

$$x \ln x + t \frac{dx}{dt} = 0.$$

By considering the derivative of $\ln(\ln x)$, solve the differential equation given $x = e$ when $t = 2$.

Express your simplified answer for x in terms of t .

- (c) In a large school, 28% of students do not play any musical instrument, 52% play exactly one musical instrument and the remainder play two or more musical instruments.

A random sample of 12 students from the school is chosen.

- (i) Find the probability that more than 9 of these students play at least one musical instrument. 2

- (ii) A random sample of 90 students from the school is now chosen. 3

Use an approximation to find the probability that fewer than 40 of these students play exactly one musical instrument.

You may wish to use the z -table provided to answer this question.

Table of values $P(Z \leq z)$ for the normal distribution $N(0, 1)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545

Question 14 continues on the following page

Question 14 (continued)

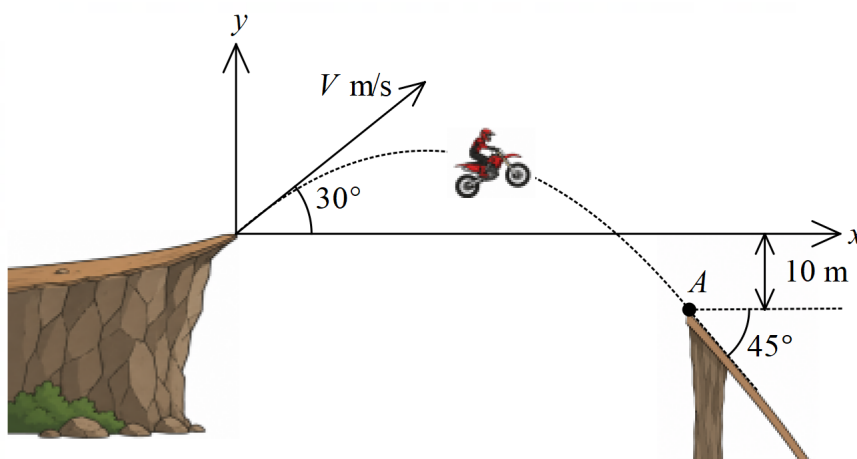
- (d) A motorcyclist attempts a gorge jump by launching from a ramp with an initial speed of $V \text{ ms}^{-1}$ at an angle of elevation of 30° to the horizontal. 4

From the initial point of launch, the position of the motorcyclist, (x, y) , at time t seconds after launch is given, by the equations of motion:

$$x = \frac{\sqrt{3}Vt}{2}$$
$$y = -5t^2 + \frac{Vt}{2}$$

You are not required to prove this.

The motorcyclist aims to land safely on a receiving ramp at point A as shown below, which is located at a vertical depth of 10 metres below the level of launch. The receiving ramp slopes downwards at a fixed angle of 45° to the horizontal.



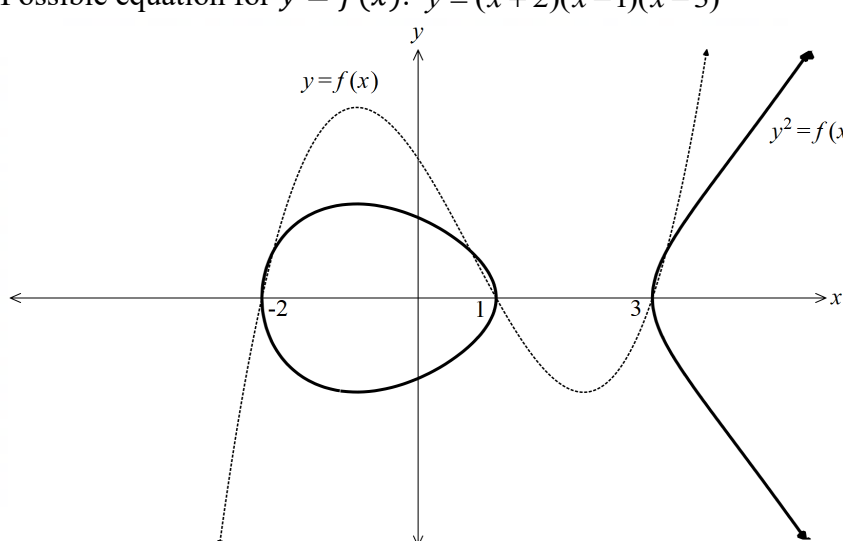
For a “perfectly smooth” landing to occur at point A , the motorcycle’s direction of motion at the exact moment of impact must be parallel to the downward slope of the receiving ramp.

Neglecting air resistance, find the exact value of the launch speed V required to successfully execute this jump.

End of paper

Section I

Award 1 mark for each correct answer.

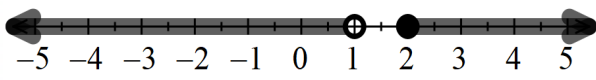
Qn	Suggested Solutions	Answer
1	$\frac{dy}{dx} = 0$ at $y = x$. Option B: $\frac{dy}{dx} = x - x = 0$ $\therefore \frac{dy}{dx} = x - y$	B
2	Selecting either 2 or 3 socks is clearly not sufficient, as you might pick one sock of each colour. Pigeonhole principle: if $n + 1$ or more objects are placed into n holes, then some hole contains at least two objects. $\therefore$ 4 socks	B
3	$\underline{a} \cdot \underline{b} = (4)(1) + 3(-1)$ $= 1$ $\text{proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{ \underline{b} ^2} \times \underline{b}$ $= \frac{1}{1^2 + (-1)^2} (\underline{i} - \underline{j})$ $= \frac{1}{2} (\underline{i} - \underline{j})$ $ \underline{i} - \underline{j} = \sqrt{1^2 + (-1)^2}$ $= \sqrt{2}$ $\therefore \hat{u} = \frac{1}{\sqrt{2}} (\underline{i} - \underline{j})$	A
4	Possible equation for $y = f(x)$: $y = (x + 2)(x - 1)(x - 3)$  Domain: requires $f(x) \geq 0$, giving $-2 \leq x \leq 1$ and $x \geq 3$ which rules out option D.	B

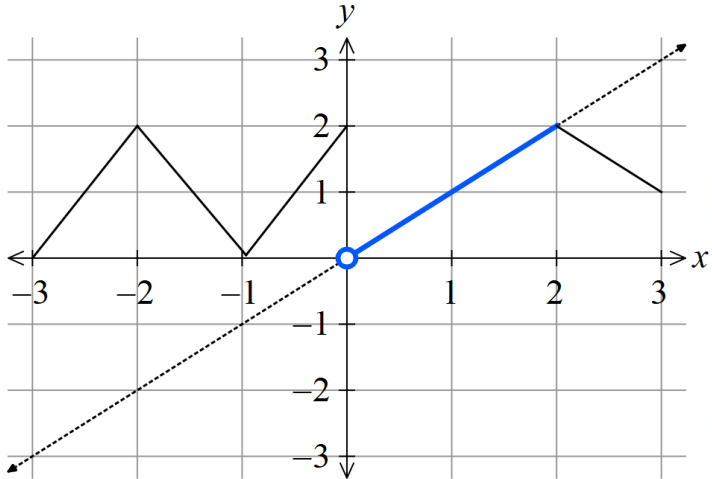
	Symmetry: $y = \pm\sqrt{f(x)}$ gives two mirrored branches about the x – axis, not y – axis. Which rules out C. Behaviour at zeroes: $\frac{dy}{dx} = \frac{f'(x)}{2\sqrt{f(x)}} \rightarrow \infty$ as $f(x) \rightarrow 0^+$, so the tangent is vertical at each zero. Since $f'(x) \neq 0$ there, the upper branch $y = +\sqrt{f(x)}$ and lower branch $y = -\sqrt{f(x)}$ meet smoothly along that shared vertical tangent, which is not a cusp. Which rules out A.	
5	$ a - b ^2 = 8$ $ a ^2 + b ^2 - 2a \cdot b = 8$ $1 + 1 - 2a \cdot b = 8$ $a \cdot b + a \cdot c = -3$	A
6	Let X be the number of defective components in a sample of n so $X \sim B(n, 0.02)$. $P(X \geq 1) = 1 - P(X = 0)$ $= 1 - 0.98^n$ $1 - 0.98^n > 0.5$ $0.98^n < 0.5$ $n \ln 0.98 < 0.5$ $n > \frac{\ln 0.5}{\ln 0.98}$ $n > 34.31$ $\therefore n = 35$	C
7	$\cos x = \frac{1 - t^2}{1 + t^2} = \frac{7}{25}$ (by Pythagoras' theorem) $25 - 25t^2 = 7 + 7t^2$ $-32t^2 = -18$ $t^2 = \frac{9}{16}$ $t = \pm \frac{3}{4}$ Since x is the first quadrant, $0 < x < \frac{\pi}{2}$, so $0 < \frac{x}{2} < \frac{\pi}{4}$ which means $t = \tan\left(\frac{x}{2}\right)$ must satisfy $0 < t < 1$. This will exclude $t = \frac{4}{3}$ (since it's greater than 1), leaving only $\frac{3}{4}$	D

8	Arranging Kaedon, Ronak and Gurjivan as a group is $3!$ ways. Arranging this group with 3 others in $4!$ ways. Once these 4 groups have been placed, Ekun has 5 places to sit. Once Ekun has seated, Inupa only has 4 choices to sit since he does not want to sit next to Ekun. Possible arrangements: $3! \times 4! \times 5 \times 4 = 2880$.	C
9	$f(x) = \sin^{-1} \left(\frac{x^2 - 3x + 2}{x^2 + 2x + 7} \right)$ $-1 \leq \frac{x^2 - 3x + 2}{x^2 + 2x + 7} \leq 1$ $\frac{x^2 - 3x + 2}{x^2 + 2x + 7} \leq 1$ Since the denominator $\frac{x^2 - 3x + 2}{x^2 + 2x + 7} \geq -1$ is positive definite, the sign does not need $x^2 - 3x + 2 \geq -x^2 - 2x - 7$ to change $2x^2 - x + 9 \geq 0$ $x^2 - 3x + 2 \leq x^2 + 2x + 7$ Since it is positive definite, $x \in \mathbb{R}$ $5x \geq -5$ $x \geq -1$ Domain $\in [-1, \infty)$	C
10	$S = \frac{1}{0!51!} + \frac{1}{1!50!} + \frac{1}{2!49!} + \dots + \frac{1}{50!1!} + \frac{1}{51!0!}$ $= \frac{1}{51!} \left(\frac{51!}{0!51!} + \frac{51!}{1!50!} + \frac{51!}{2!49!} + \dots + \frac{51!}{50!1!} + \frac{51!}{51!0!} \right)$ $= \frac{1}{51!} \left({}^{51}C_0 + {}^{51}C_1 + {}^{51}C_2 + \dots + {}^{51}C_{50} + {}^{51}C_{51} \right)$ $\because {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$ $\therefore S = \frac{2^{51}}{51!}$	B

Section II

Q	Suggested Solutions	Marking Criteria
11(a)	$x = \cos 2t$ $x = 2 \cos^2 t - 1$ $\frac{x+1}{2} = \cos^2 t$ $\therefore y = \cos t$ $y^2 = \cos^2 t$ $\therefore y^2 = \frac{x+1}{2} \text{ or } 2y^2 = x+1$ $\text{or } 2y^2 - 1 = x \text{ for } x \in [-1, 1]$	2 marks - Correct solution 1 mark - Correctly rearranges to find $\cos^2 t$ or equivalent merit Note: Do not deduct marks for not writing the restriction
11(b)	$\text{General term } T_{r+1} = \binom{8}{r} (1)^{8-r} (-x)^r$ $= \binom{8}{r} (-1)^r x^r$ $\text{Binomial expansion} = (1+2x) \times \sum_{r=0}^8 \binom{8}{r} (-1)^r x^r$ $\text{When } r=4, 1 \times \binom{8}{4} (-1)^4 x^4$ $= 70x^4$ $\text{When } r=3, 2x \times \binom{8}{3} (-1)^3 x^3$ $= -112x^4$ $\therefore 70x^4 + (-112)x^4 = -42x^4$ $\therefore \text{the coefficient of } x^4 \text{ is } -42$	2 marks - Correct solution 1 mark - Finds one correct coefficient (70 or -112)
11(c)	$\int \frac{2}{\sqrt{4-25x^2}} dx = \frac{2}{5} \int \frac{1}{\sqrt{\frac{4}{25} - x^2}} dx$ $= \frac{2}{5} \sin^{-1} \left(\frac{5x}{2} \right) + C$	2 marks - Correct solution 1 mark - Obtains a primitive in the form of $k \sin^{-1} \left(\frac{5x}{2} \right)$ or $k \cos^{-1} \left(\frac{5x}{2} \right)$

11(d)	$\frac{x+4}{x-1} \leq 6, x \neq 1$ $\frac{x+4-6(x-1)}{x-1} \leq 0$ $\frac{x+4-6x+6}{x-1} \leq 0$ $\frac{-5x+10}{x-1} \leq 0$ $\frac{-5(x-2)}{x-1} \leq 0$ $\frac{x-2}{x-1} \leq 0$ Testing critical points  $\therefore x < 1 \text{ or } x \geq 2$	3 marks - Correct solution 2 marks - Bald answer - Identifies the correct critical points via a correct method - Correct conclusion to their critical points obtained using a correct method 1 mark - Uses a correct method - Acknowledges a problem with the denominator
11(ei)	Since $\overrightarrow{AM} = \frac{3}{4} \overrightarrow{AB}$ and $\overrightarrow{AB} = \vec{b} - \vec{a}$ $\overrightarrow{OM} = \overrightarrow{OA} + \overrightarrow{AM}$ $\overrightarrow{OM} = \vec{a} + \frac{3}{4}(\vec{b} - \vec{a})$ $\overrightarrow{OM} = \frac{1}{4}\vec{a} + \frac{3}{4}\vec{b}$ Since $\overrightarrow{OC} = 4\vec{b} - 3\vec{a}$ $\overrightarrow{MC} = \overrightarrow{OC} - \overrightarrow{OM}$ $= (4\vec{b} - 3\vec{a}) - \left(\frac{1}{4}\vec{a} + \frac{3}{4}\vec{b}\right)$ $= \frac{13}{4}\vec{b} - \frac{13}{4}\vec{a}$ $\overrightarrow{AM} = \frac{3}{4}(\vec{b} - \vec{a})$ $= \frac{3}{13} \left(\frac{13}{4}\vec{b} - \frac{13}{4}\vec{a} \right)$ $= \frac{3}{13} \overrightarrow{MC}$ Since $\overrightarrow{AM}$ is a scalar multiple of $\overrightarrow{MC}$, $\overrightarrow{AM} \parallel \overrightarrow{MC}$. As $\overrightarrow{AM}$ and $\overrightarrow{MC}$ share a common point M, therefore A, M and C are collinear.	2 marks - Correct solution 1 mark - Correctly finds $\overrightarrow{OM}$, $\overrightarrow{AC}$ or $\overrightarrow{MC}$ in terms of $\vec{a}$ and $\vec{b}$

11(eii)	$\overrightarrow{AM} = \frac{3}{13} \overrightarrow{MC}$ $\frac{ \overrightarrow{AM} }{ \overrightarrow{MC} } = \frac{3}{13}$ $AM : MC = 3 : 13$	1 mark - Correct solution
11(f)	Show true for $n = 1$, $2^1 + 6^1 = 2 + 6$ $= 8, \text{ clearly divisible by } 8$ $\therefore \text{ true for } n = 1$ Assume true for $n = k, k \in \mathbb{Z}$ $2^k + 6^k = 8M, M \in \mathbb{Z}$ Show true for $n = k + 1, k \in \mathbb{Z}$ $2^{k+1} + 6^{k+1} = 8N, N \in \mathbb{Z}$ $2^{k+1} + 6^{k+1}$ $= 2^k \times 2 + 6^k \times 6$ $= (8M - 6^k) \times 2 + 6^{k+1}$ $= 8M \times 2 - 6^k(2) + 6^k \times 6$ $= 8M \times 2 + 6^k \times 4$ $= 8M \times 2 + 4 \times 3^k \times 2^k$ $= 8M \times 2 + 4 \times 2 \times 2^{k-1} \times 3^k \quad (k-1 \geq 0)$ $= 8(2M + 2^{k-1} \times 3^k)$ $= 8N, \text{ where } N = 2M + 2^{k-1} \times 3^k, \text{ and } N \text{ is an integer}$ $\therefore \text{ true for } n = k + 1, \text{ if true for } n = k$ $\therefore \text{ Since true for } n = 1 \text{ by mathematical induction,}$ $\text{true for all positive integers } n$	There are 4 key parts of the induction; 1. Proving the result true for $n = 1$ 2. Clearly stating the assumption and what is to be proven 3. Using the assumption in the proof 4. Correctly proving the required statement 3 marks - Successfully does all the 4 keys parts 2 marks - Successfully does 3 of the 4 key parts 1 mark - Successfully does 2 of the 4 key parts
12 (ai)	 From the graph: $0 < x \leq 2$	2 marks - Correct solution 1 mark - Correctly identifies $x = 2$ in their solution or sketches $f(x)$ or equivalent merit

12(aii)		2 marks - Correct graph 1 mark - Correctly identifies the vertical asymptotes or - Correctly identifies the turning points
12(b)	$\int_0^{\frac{\pi}{4}} \frac{\tan x \sec^2 x}{(1 + \tan^2 x)^2} dx = \frac{1}{2} \int_1^2 \frac{1}{u^2} du$ $= \frac{1}{2} \left[-\frac{1}{u} \right]_1^2 = \frac{1}{2} \left(-\frac{1}{2} + \frac{1}{1} \right) = \frac{1}{4}$ $u = 1 + \tan^2 x$ $u = \sec^2 x$ $\frac{du}{dx} = 2 \sec^2 x \tan x$ $du = 2 \sec^2 x \tan x dx$ $x = 0, u = 1$ $x = \frac{\pi}{4}, u = 2$	3 marks - Correct solution 2 marks - Obtains correct primitive in terms of u or equivalent merit 1 mark - Obtains correct integrand in terms of u or equivalent merit
12(ci)	$P(x) = x^3 - 5x^2 + 8x + k$ $P'(x) = 3x^2 - 10x + 8$ $P'(\alpha) = 3\alpha^2 - 10\alpha + 8 = 0$ (since $P(x)$ has a root of multiplicity 2). $(\alpha - 2)(3\alpha - 4) = 0$ $\therefore \alpha = 2$ or $\frac{4}{3}$	2 marks - Correct solution 1 mark - Shows understanding that $P'(\alpha) = 0$ or equivalent merit
12(cii)	If α is a root then $P(\alpha) = 0$, Test $\alpha = 2$, $P(2) = 2^3 - 5(2)^2 + 8(2) + k = 0$ $\therefore k = -4$ Since k is an integer, $\therefore k = -4$ Test $\alpha = \frac{4}{3}$, $P\left(\frac{4}{3}\right) = \left(\frac{4}{3}\right)^3 - 5\left(\frac{4}{3}\right)^2 + 8\left(\frac{4}{3}\right) + k = 0$ $\therefore k = -\frac{112}{27}$	2 marks - Correct solution 1 mark - Uses $P(2)$ to find the value of k Note: do not need to test $\alpha = \frac{4}{3}$ if explanation provided

12(di)	$\cos 3x \cos x \equiv \frac{1}{2}(\cos 2x + \cos 4x)$ $LHS = \cos 3x \cos x$ $= \frac{1}{2}[\cos(3x - x) + \cos(3x + x)]$ $= \frac{1}{2}(\cos 2x + \cos 4x)$ $= RHS$	1 mark - Correct proof
12(dii)	$4 \cos x \cos 2x \cos 3x = 1$ $4 \cos 2x \left[\frac{1}{2}(\cos 4x + \cos 2x) \right] = 1$ $2 \cos 2x(\cos 4x + \cos 2x) = 1$ $2 \cos 2x(2 \cos^2 2x + \cos 2x - 1) = 1$ $4 \cos^3 2x + 2 \cos^2 2x - 2 \cos 2x = 1$ $4 \cos^3 2x + 2 \cos^2 2x - 2 \cos 2x - 1 = 0$ $(2 \cos^2 2x - 1)(2 \cos 2x + 1) = 0$ $2 \cos^2 2x - 1 = 0$ $\cos^2 2x = \frac{1}{2}$ $\cos 2x = \pm \frac{1}{\sqrt{2}}$ $2x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ $x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$ $\therefore x = \frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$ $2 \cos 2x + 1 = 0$ $\cos 2x = -\frac{1}{2}$ $2x = \frac{2\pi}{3}, \frac{4\pi}{3}$ $x = \frac{\pi}{3}, \frac{2\pi}{3}$ $\cos 4x = \cos 2(2x)$ $= 2 \cos^2 2x - 1$ Grouping by pairs	3 marks - Correct solution 2 marks - Correctly finds 2 values for x . 1 mark - Correctly forms a simplified equation in terms of $\cos 2x$.

13(a)	$\sin^{-1} x$ is normally defined for $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ but for $\frac{\pi}{2} < y < \pi$, we need to use the identity $\sin(\pi - \theta) = \sin \theta$ so that our interval works. $\therefore \sin(\pi - y) = \frac{x}{1+x^2}$ $\pi - y = \sin^{-1} \frac{x}{1+x^2}$ $y = \pi - \sin^{-1} \frac{x}{1+x^2}$ $\frac{dy}{dx} = \frac{-\left(\frac{1-x^2}{(1+x^2)^2}\right)}{\sqrt{1-\left(\frac{x}{1+x^2}\right)^2}}$ $= -\left(\frac{1-x^2}{(1+x^2)^2}\right) \div \sqrt{\frac{1+2x^2+x^4-x^2}{(1+x^2)^2}}$ $= -\left(\frac{1-x^2}{(1+x^2)^2}\right) \times \frac{1+x^2}{\sqrt{1+x^2+x^4}}$ $= \frac{x^2-1}{(1+x^2)\sqrt{1+x^2+x^4}}$ $f(x) = \frac{x}{1+x^2}$ $f'(x) = \frac{1+x^2-2x^2}{(1+x^2)^2}$ $= \frac{1-x^2}{(1+x^2)^2}$	3 marks - Correct proof 2 marks - Correctly derives $\sin^{-1} \frac{x}{1+x^2}$ 1 mark - Correctly uses the identity to derive sin inverse or equivalent merit
13(b)	$V = \pi \int_0^{\frac{\pi}{6}} \left(\sqrt{\sin 2x + \sin^2 2x}\right)^2 dx$ $= \pi \int_0^{\frac{\pi}{6}} \sin 2x + \sin^2 2x dx$ $= \pi \left[\int_0^{\frac{\pi}{6}} \sin 2x dx + \int_0^{\frac{\pi}{6}} \sin^2 2x dx \right]$ $= \pi \left[\left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}} + \int_0^{\frac{\pi}{6}} \frac{1}{2} - \frac{1}{2} \cos 4x dx \right]$ $= \pi \left[-\frac{1}{2} \cos 2x + \frac{x}{2} - \frac{1}{8} \sin 4x \right]_0^{\frac{\pi}{6}}$ $= \pi \left(-\frac{1}{2} \cos 2\left(\frac{\pi}{6}\right) + \frac{\pi}{2(6)} - \frac{1}{8} \sin 4\left(\frac{\pi}{6}\right) + \frac{1}{2} \cos 4(0) - \frac{0}{2} + \frac{1}{8} \sin 4(0) \right)$ $= \pi \left(-\frac{1}{4} + \frac{\pi}{12} - \frac{\sqrt{3}}{16} + \frac{1}{2} \right)$ $= \pi \left(\frac{1}{4} + \frac{\pi}{12} - \frac{\sqrt{3}}{16} \right) \text{ cubic units.}$	3 marks - Correct solution 2 marks - Obtains correct primitive 1 mark - Correctly uses the double angle formula for $\sin^2 2x$

13(c)	if $\underline{u} \perp \underline{v} \Rightarrow (\underline{a} + 3\underline{b}) \cdot (7\underline{a} - 5\underline{b}) = 0$ $7 \underline{a} ^2 - 15 \underline{b} ^2 + 16\underline{a} \cdot \underline{b} = 0 \quad (1)$ if $\underline{z} \perp \underline{w} \Rightarrow (\underline{a} - 4\underline{b}) \cdot (7\underline{a} - 2\underline{b}) = 0$ $7 \underline{a} ^2 + 8 \underline{b} ^2 - 30\underline{a} \cdot \underline{b} = 0 \quad (2)$ Solve simultaneously $30 \times (1) = 16 \times (2)$ $210 \underline{a} ^2 - 450 \underline{b} ^2 + 480\underline{a} \cdot \underline{b} = 112 \underline{a} ^2 + 128 \underline{b} ^2 - 480\underline{a} \cdot \underline{b}$ $322 \underline{a} ^2 = 322 \underline{b} ^2$ $\therefore \underline{a} = \underline{b}$ From equation (2) $15 \underline{a} = 30\underline{a} \cdot \underline{b}$ since $\underline{a} \cdot \underline{b} = \underline{a} \underline{b} \cos \theta$ $15 \underline{a} ^2 = 30 \underline{a} \underline{b} \cos \theta$ $\cos \theta = \frac{1}{2}$ $\therefore \theta = 60^\circ$	4 marks - Correct solution 3 marks - Correctly obtains $\underline{a} = \underline{b}$ or equivalent merit 2 marks - Forms a pair of simultaneous equations in terms of $\underline{a}$ and $\underline{b}$ or equivalent merit 1 mark - Uses $\underline{u} \cdot \underline{v} = 0$ or equivalent
13(di)	$\frac{dH}{dt} = kH \left(1 - \frac{H}{1000} \right)$ $\frac{dH}{dt} = kH \left(\frac{1000 - H}{1000} \right)$ $\int_{100}^H \frac{1}{H} + \frac{1}{1000 - H} dH = \int_0^t k dt$ $\left[\ln \frac{H}{1000 - H} \right]_{100}^H = [kt]_0^t \quad (\text{since } 100 \leq H \leq 1000)$ $\ln \left(\frac{H}{1000 - H} \right) - \ln \left(\frac{100}{1000 - 100} \right) = kt - k(0)$ $\ln \left(\frac{H}{1000 - H} \times \frac{900}{100} \right) = kt$ $\ln \left(\frac{9H}{1000 - H} \right) = kt$ $e^{kt} = \frac{9H}{1000 - H}$ $1000e^{kt} = 9H + He^{kt}$ $H = \frac{1000e^{kt}}{9 + e^{kt}}$ $\therefore H = \frac{1000}{1 + 9e^{-kt}}$	3 marks - Correct solution 2 marks - Establishes $kt = \ln \frac{9H}{1000 - H}$ or equivalent merit 1 mark - Uses the given result to find the primitive of two log functions

13(dii)	When $t = 2, H = 134$ $134 = \frac{1000}{1 + 9e^{-k(2)}}$ $1 + 9e^{-2k} = \frac{1000}{134}$ $e^{-2k} = \frac{433}{603}$ $-2k = \ln\left(\frac{433}{603}\right)$ $k = -\frac{1}{2}\ln\left(\frac{433}{603}\right)$ ≈ 0.1656	2 marks - Correct solution 1 mark - Obtains $e^{-2k} = \frac{433}{603}$ or equivalent merit
14(a)	$\frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt}$ $\frac{dr}{dt} = \frac{1}{\pi r \times 0.006} \times 0.0008$ $= \frac{2}{15\pi(3)}$ $= \frac{2}{45\pi}$ $V = \pi r^2 h$ $= \pi r^2 \times 0.003$ $\frac{dV}{dr} = \pi r \times 0.006$	2 marks - Correct solution 1 mark - Finds correct expression for $\frac{dr}{dt}$ in terms of r. - Finds a correct expression for $\frac{dV}{dr}$
14(bii)	$x \ln x + t \frac{dx}{dt} = 0$ $\frac{dx}{dt} = \frac{-x \ln x}{t}$ $\int \frac{1}{x \ln x} dx = - \int \frac{1}{t} dt$ $\ln(\ln x) = -\ln t + c$ when $x = e, t = 2$ $\ln(\ln e) = -\ln 2 + c$ $0 = -\ln 2 + c$ $\therefore c = \ln 2$ $\ln(\ln x) = -\ln t + \ln 2$ $\ln x = e^{-\ln t + \ln 2}$ $\ln x = 2t^{-1}, \text{ since } t > 0$ $x = e^{\frac{2}{t}}$ $\left \frac{d}{dx}(\ln(\ln(x))) = \frac{1}{x \ln x} \right$	4 marks - Correct solution 3 marks - Correctly finds the value of c, or equivalent merit 2 marks - Correctly integrates the variables 1 mark - Finds the derivative of $\ln(\ln(x))$
14(ci)	72% play 1 instrument and 2 or more instruments $M \sim B(12, 0.72)$ $P(M > 9) = P(10) + P(11) + P(12)$ $= {}^{12}C_{10} \times 0.72^{10} \times 0.28^2 + {}^{12}C_{11} \times 0.72^{11} \times 0.28^1 + {}^{12}C_{12} \times 0.72^{12} \times 0.28^0$ $= 0.304$	2 marks - Correct solution 1 mark - Recognises 72% play 1 instrument and 2 or more instruments

14(cii)	Method 1 (applying continuity correction): $X \sim B(90, 0.52)$ $\mu = 90 \times 0.52 \quad \sigma^2 = 46.8 \times 0.48$ $= 46.8 \quad \sigma = \sqrt{22.464}$ $P(X < 40) \rightarrow P(X < 39.5)$ $P\left(Z < \frac{39.5 - 46.8}{\sqrt{22.464}}\right) = P(Z < -1.5402)$ $= 1 - \phi(1.540)$ $= 1 - 0.9382$ $= 0.0618$ Method 2 (without continuity correction): $P(X < 40)$ $P(Z < 40) = P\left(Z < \frac{40 - 46.8}{\sqrt{22.464}}\right)$ $= P(Z < -1.434714...)$ $= 1 - \phi(1.434714...)$ $= 1 - 0.9236$ $= 0.0764$ Method 3 (sample proportion): $P\left(\hat{p} < \frac{40}{90}\right) = P\left(\frac{\frac{4}{9} - 0.52}{0.0526}\right) \quad \left \begin{array}{l} \bar{x} = 0.52 \\ \sigma = \sqrt{\frac{0.52 \times 0.48}{90}} \end{array} \right.$ $= P(-1.4364...) = 0.0526$ $= 1 - \phi(1.4364...)$ $= 1 - 0.9251$ $= 0.0749$ Note: Students were not deducted a mark for using $P(X \leq 39)$	3 marks - Correct solution 2 marks - Obtains correct z score 1 mark - Obtains correct standard deviation
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14(d)

$$\frac{dx}{dt} = \frac{\sqrt{3}V}{2} \quad \frac{dy}{dt} = -10t + \frac{V}{2}$$

$$\frac{dx}{dt} \times \frac{dt}{dy} = \left(-10t + \frac{V}{2}\right) \times \frac{2}{\sqrt{3}} V$$

$$\frac{dx}{dy} = \frac{-20t + V}{\sqrt{3}V}$$

$$-1 = \frac{-20t + V}{\sqrt{3}V}$$

$$-\sqrt{3}V = -20t + V$$

$$t = \frac{\sqrt{3}+1}{20} V$$

Bike hits the ground when $y = -10$

$$-5t^2 + \frac{Vt}{2} = -10$$

$$-10t^2 + Vt = -20$$

$$Vt = 10t^2 - 20$$

$$V = 10t - \frac{20}{t}$$

$$V = \frac{\sqrt{3}+1}{2} V - \frac{400}{(\sqrt{3}+1)V}$$

$$\frac{\sqrt{3}-1}{2} V = \frac{400}{(\sqrt{3}+1)V}$$

$$V^2 = \frac{800}{(\sqrt{3}+1)(\sqrt{3}-1)}$$

$$= 400$$

$$V = \pm 20$$

$\therefore$ bike is launched at 20m/s

4 marks

- Correct solution

3 marks

- Use the given information to find an equation in terms of V

2 marks

- Finds t in terms of V

1 mark

- Establishes that $\frac{dy}{dx} = -1$ on impact